

ROXY TABLET 150 MG

DESCRIPTION
White round biconvex film-coated tablet, embossed with “Roxy” and “SPS” on one side and scored on the other side. Each film-coated tablet contains Roxithromycin 150 mg.

INDICATION
Roxithromycin is indicated for:

- Upper respiratory tract infections, e.g. acute pharyngitis and sinusitis.
- Lower respiratory tract infections, e.g. pneumonia, bronchitis, atypical pneumonia.
- Skin and skin structural infections.
- Non-gonococcal urethritis.
- Prophylaxis of meningococcal meningitis subjects who have come in contact with the patients.

RECOMMENDED DOSAGE
Adults:
For most infections, oral roxithromycin is effective in doses of 150 mg twice a day or 300 mg once a day. Therapy should be continued for at least 10 days. The maximum duration of therapy is 4 weeks. Doses should be taken in the morning and evening before meals.

CONTRAINDICATIONS

- Hypersensitivity to roxithromycin, macrolides or to any of the excipients.
- Severely impaired hepatic function.
- Concomitant therapy with vasoconstrictive ergot alkaloids.

WARNINGS AND PRECAUTIONS
Roxithromycin, like erythromycin, has been shown *in vitro* to elicit a concentration-dependent lengthening in cardiac action potential duration. Such an effect is manifested only at supratherapeutic concentrations. Accordingly, the recommended doses should not be exceeded.

Prolonged or repeated use of antibiotics including roxithromycin may result in superinfection by resistant organisms. In the event of superinfection, roxithromycin should be discontinued and appropriate therapy instituted. When indicated, incision, drainage or other appropriate surgical procedures should be performed in conjunction with antibiotic therapy.

Pseudomembranous colitis
Antibiotic associated pseudomembranous colitis has been reported with many antibiotics. A toxin produced by *Clostridium difficile* appears to be the primary cause. If pseudomembranous colitis is suspected, roxithromycin must be stopped immediately.

The severity of the colitis may range from mild to life threatening. It is important to consider this diagnosis in patients who develop diarrhoea particularly if severe, persistent and/or bloody, or colitis in association with antibiotic use (this may occur during treatment and up to several weeks after cessation of antibiotic therapy). Mild cases usually respond to drug discontinuation alone. However, in moderate to severe cases, appropriate therapy with a suitable oral antibacterial agent effective against *Clostridium difficile* should be considered. Fluids, electrolytes and protein replacement therapy should be provided when indicated. Drugs that delay peristalsis, e.g. opiates and diphenoxylate with atropine, may prolong and/or worsen the condition and should not be used.

Diarrhoea, particularly if severe, persistent and/or bloody, during or after treatment with roxithromycin, may be symptomatic of pseudomembranous colitis. If pseudomembranous colitis is suspected, roxithromycin must be stopped immediately.

QT-prolongation
Macrolides, including roxithromycin, have the potential to prolong the QT interval in certain conditions. Therefore, roxithromycin should be used with caution in patients with congenital prolongation of the QT interval, with ongoing proarrhythmic conditions (i.e. uncorrected hypokalaemia or hypomagnesaemia, clinically significant bradycardia), and in patients receiving Class IA and III antiarrhythmic agents and drugs such as astemizole, cisapride or pimozide.

Myasthenia gravis
As with other macrolides, roxithromycin may have the potential to aggravate the myasthenia gravis. Patients with myasthenia gravis taking roxithromycin should be advised to immediately seek medical attention if they experience exacerbation of their symptoms. Roxithromycin must then be discontinued and supportive care administered as medically indicated.

Hypersensitivity reactions
In the event of severe acute hypersensitivity reactions, such as anaphylaxis, severe cutaneous adverse reactions (SCARs) [e.g. Stevens-Johnson syndrome (SJS), toxic epidermal necrolysis (TEN), drug reaction with eosinophilia and systemic symptoms (DRESS)] and acute generalized exanthematous pustulosis (AGEP)], roxithromycin should be discontinued immediately and appropriate treatment should be urgently initiated.

Ergotism
Severe vasoconstriction (“ergotism”) with possibly necrosis of the extremities has been reported when macrolides antibiotics have been associated with vasoconstrictive ergot alkaloids. Absence of treatment by these alkaloids must always be checked before prescribing roxithromycin.

Use in hepatic impairment
The safety of roxithromycin has not been demonstrated in patients with impaired hepatic function. Caution should be exercised if roxithromycin is administered to patients with impaired hepatic function. If administered to patients with severely impaired hepatic function (e.g. hepatic cirrhosis with jaundice and/or ascites), consideration should be given to reducing the daily dosage to half the usual dosage.

Use in renal impairment
The safety of roxithromycin has not been demonstrated in patients with impaired renal function. Caution should be exercised if roxithromycin is administered to patients with impaired renal function. Renal excretion of roxithromycin and its metabolites accounts for a small percentage of an oral dose. The dosage should be kept unchanged in renal insufficiency.

Use in elderly
No dosage adjustment is required in elderly patients.

Use in children
The maintenance of therapeutic safety margins is primarily dependent on high affinity binding of roxithromycin to plasma alpha-1-acid glycoprotein and will be compromised by any circumstances attenuating the extent of this binding. It is recommended that the approved paediatric dosage regimen (i.e. 5 to 8 mg/kg/day for a maximum of ten days) be adhered to strictly.

Neutropenia was observed in children treated with roxithromycin. It is not known whether this is an effect of the drug, or whether it reflects a normal fluctuation of the neutrophil count or a response to infection in children.

Effects on Ability to Drive and Use Machine
Attention should be drawn to the possibility of dizziness, visual impairment and blurred vision.

DRUG INTERACTIONS
Roxithromycin is a weak CYP3A inhibitor. The effect of roxithromycin on exposure to drugs predominantly cleared by CYP3A metabolism would be expected to be 2-fold or less. Caution should be exercised when roxithromycin is concomitantly prescribed with drugs metabolised by CYP3A (e.g. rifabutin and bromocriptine).

Roxithromycin has a much lower affinity for cytochrome P450 than erythromycin, and consequently has fewer interactions. Interactions may be observed, however, with drugs that bind to alpha-1-acid glycoprotein, e.g. disopyramide.

Roxithromycin does not appear to interact with oral contraceptives containing oestrogens and progestogens, prednisolone, carbamazepine, ranitidine or antacids.

Ergot alkaloids: Concomitant administration of roxithromycin and ergot alkaloid derivatives (e.g. ergotamine and dihydroergotamine) could lead to severe vasoconstriction (“ergotism”) with possibly necrosis of the extremities. Administration of roxithromycin to patients taking ergot alkaloids is contraindicated. Absence of treatment with these alkaloids must always be checked before prescribing roxithromycin.

Theophylline: A study in normal subjects concurrently administered roxithromycin and theophylline has shown some increase in the plasma concentration of the latter. While a change in dosage is usually not required, patients with high levels of theophylline at commencement of treatment should have levels monitored.

Disopyramide: An *in vitro* study has shown that roxithromycin can displace protein bound disopyramide. Such an effect *in vivo* could result in increased serum levels of disopyramide. Consequently ECG and, if possible, disopyramide serum levels should be monitored.

Terfenadine: Some macrolide antibiotics may increase serum levels of terfenadine. This can result in severe cardiovascular adverse events, including QT prolongation, torsades de pointes and other ventricular arrhythmias. Although this kind of reaction has not been demonstrated with roxithromycin and studies with a limited number of healthy volunteers have shown no pharmacokinetic interaction or relevant ECG-changes, the combination of roxithromycin and terfenadine is contraindicated.

Astemizole, cisapride, pimozide: Other drugs, such as astemizole, cisapride or pimozide, which are metabolised by the hepatic isozyme CYP3A4, have been associated with QT interval prolongation and/or cardiac arrhythmias (typically torsades de pointes) as a result of an increase in their serum level subsequent to interaction with significant inhibitors of this isozyme, including some macrolide antibacterials. Although roxithromycin has no or limited ability to complex CYP3A4 and therefore to inhibit the metabolism of other drugs processed by this isozyme, a potential for clinical interaction of roxithromycin with the above mentioned drugs cannot be either ascertained or ruled out in confidence; therefore, concomitant administration of roxithromycin and such drugs is not recommended.

Vitamin K antagonists: Increases in prothrombin time (international normalised ratio (INR)) have been reported in patients treated concomitantly with roxithromycin and warfarin or the related vitamin K antagonist phenprocoumon, and severe bleeding episodes have occurred as a consequence. INR should be monitored during combined treatment with roxithromycin and Vitamin K antagonists.

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Digoxin and other cardiac glycosides: Roxithromycin may increase the absorption of digoxin. This effect, common to other macrolides, may very rarely result in cardiac glycoside toxicity. This may be manifested by symptoms such as nausea, vomiting, diarrhoea, headache or dizziness; cardiac glycoside toxicity may also elicit heart conduction and/or rhythm disorders. Consequently, in patients treated with roxithromycin and digoxin or another cardiac glycoside, ECG and, if possible, the serum level of the cardiac glycoside should be monitored; this is mandatory if symptoms which may suggest cardiac glycoside overdosage occur. Roxithromycin should be used with caution in patients receiving Class IA and Class III antiarrhythmic agents.

HMG-CoA reductase inhibitors: When roxithromycin and an HMG-CoA reductase inhibitor (statin) are combined, there is a potential risk of muscle related adverse events, such as rhabdomyolysis due to a possible increase of the statin exposure. Caution should be exercised when a statin is combined with roxithromycin and patients should be monitored for signs and symptoms of myopathy.

Midazolam: Roxithromycin may increase the area under the midazolam concentration-time curve (AUC) and the midazolam half-life; therefore, the effects of midazolam may be enhanced and prolonged in patients treated with roxithromycin. There is no conclusive evidence for an interaction between roxithromycin and triazolam.

Bromocriptine: Roxithromycin may increase the AUC and plasma concentrations of bromocriptine, which could lead to an increased risk for adverse effects of the compound.

Rifabutin: Roxithromycin can increase the plasma concentration of rifabutin.

Theophylline and Cyclosporin: A slight increase in plasma concentrations of theophylline or cyclosporin has been observed. This does not generally necessitate altering the usual dosage.

PREGNANCY AND LACTATION

Pregnancy

The safety of roxithromycin for the human foetus has not been established. Roxithromycin should not be used during pregnancy unless clearly indicated.

Lactation

Small amounts of roxithromycin are excreted in the breast milk. Breast-feeding or treatment of the mother should be discontinued as necessary.

SIDE EFFECTS

Blood and lymphatic system disorders: Eosinophilia, agranulocytosis, neutropenia, thrombocytopenia

Infections and infestations: Superinfection with resistant bacteria or fungi on prolonged use, *Clostridium difficile* colitis (pseudomembranous colitis), moniliasis (candidiasis)

Repeated evaluation of the patient's condition is essential. In the event of superinfection, appropriate measures should be taken.

Immune system disorders: Angioedema, anaphylactic shock

Psychiatric disorders: Confusion, hallucination, psychosis

Nervous system disorders: Headache, dizziness, disorders of taste and/or smell, paraesthesia

Eye disorders: Visual impairment, blurred vision

Ear and labyrinth disorders: Temporary deafness, hypacusis, vertigo, tinnitus

Cardiac disorders: QT interval prolongation, ventricular tachycardia, Torsade de pointes

Respiratory, thoracic and mediastinal disorders: Bronchospasm

Gastrointestinal disorder: Nausea, vomiting, epigastric pain (dyspepsia), diarrhoea (sometimes containing blood), obstipation, anorexia, flatulence, pseudomembranous colitis. Symptoms of pancreatitis have been observed; most patients had received other drugs for which pancreatitis is a known adverse reaction.

Hepatobiliary disorders: Cholestatic hepatitis, hepatocellular acute hepatitis, jaundice

Skin and subcutaneous tissue disorders: Rash, redness, urticaria, erythema multiforme, eczema, pruritus, purpura.
Frequency not known: Severe cutaneous adverse reactions (SCARs) including Stevens-Johnson syndrome (SJS), toxic epidermal necrolysis (TEN), drug reaction with eosinophilia and systemic symptoms (DRESS) & acute generalised exanthematous pustulosis (AGEP).

Musculoskeletal and connective tissue disorder: Myasthenia gravis

Investigations: Moderate increases in serum transaminases (AST and ALT) and/or alkaline phosphatase levels have been observed and are somewhat more likely to occur in the elderly (> 65 years).

OVERDOSAGE

In cases of overdose, gastric lavage, other measures to limit GI absorption, and symptomatic treatment should be initiated without delay. There is no specific antidote.

PHARMACODYNAMICS

Roxithromycin is a semi-synthetic macrolide antibiotic with a broad and essentially bacteriostatic action against many Gram-positive and to a lesser extent some Gram-negative bacteria, as well as other organisms including *mycoplasmas*, *spirochaetes*, *chlamydiae* and *rickettsiae*. Roxithromycin inhibits RNA-dependent protein synthesis at the chain elongation step in susceptible organisms. It reversibly attaches to the P site on the 50S subunit of the bacterial ribosome, resulting in blockage of the transpeptidation or translocation reactions, inhibition of protein synthesis and hence inhibition of cell growth. Its action is predominantly bacteriostatic, but high concentration is slowly bactericidal against the more sensitive strains. Intracellular penetration is an important contributor to the antimicrobial activity of roxithromycin, which concentrates into lymphocytes and macrophages at concentrations which far exceed those in the plasma. This is advantageous in the eradication of certain intracellular pathogens such as Chlamydia, Legionella, Listeria and Toxoplasma, and there has been some interest in their potential synergy with host defence mechanisms *in vivo*. Its actions are increased at moderately alkaline pH (up to about 8.5) particularly in Gram-negative species, probably because of the improved cellular penetration of the non-ionised form of the drug. Roxithromycin augments the adhesive and chemotactic activity or polymorphonuclear neutrophils, the basic functions that enables these cells to reach the site of infection; it also enhances phagocytosis and bacterial lysis.

Microbiology:

Antibacterial Activity: Roxithromycin is active against the majority of significant respiratory pathogens, including *Streptococcus pyogenes* group A, group B, C, F & G, *Streptococci*, and *S. pneumoniae*. Organism implicated in bacterial endocarditis such as *S. miteor*, *S. mutans*, and *S. sanguis* are also inhibited by low concentrations of roxithromycin. Other susceptible pathogens include *Bacillus cereus*, *Bordetella pertussis*, *Chlamydia trachomatis*, *Corynebacterium diphtheriae*, *Gardnerella vaginalis*, *Haemophilus ducreyi*, *H. influenzae*, *Helicobacter jejuni*, *Legionella pneumophila*, *Listeria monocytogenes*, *Moraxella (Branhamella) catarrhalis*, *Mycobacterium chelonae*, *M. tuberculosis*, *Mycoplasma pneumoniae*, *Pasteurella multocida*, *Staphylococcus aureus*, *S. epidermidis* and *Ureaplasma urealyticum*. Activity against anaerobes such as *Bacteroides fragilis* is variable to poor, whereas other species such as *Actinomyces spp.*, *Bacteroides oralis*, *B. melaninogenicus*, *B. ureolyticus*, *Bifidobacterium spp.*, *Eubacterium spp.*, *Peptostreptococcus spp.*, and *Propionibacterium acnes* are susceptible.

Gram-negative organisms such as *Pseudomonas* and the *Enterobacteriaceae* are uniformly resistant, as are organisms exhibiting plasmid-mediated resistance to erythromycin.

PHARMACOKINETICS

Absorption: Roxithromycin is well and rapidly absorbed from GI tract. It produces significantly higher serum levels, is better absorbed and is more stable against hydrolysis by gastric acid than erythromycin. Peak plasma levels of about 6 to 8 µg per ml are reached about 2 to 2.2 hours after a single dose of 150 mg. Absorption is reduced when taken with food. 2 tablets taken at 12 hours intervals can ensure 24 hours effective plasma concentrations against sensitive organisms.

Distribution: It is widely distributed in tissue and body fluid. Roxithromycin is mainly bound to α-1 acid glycoprotein; it is 96% bound at through concentrations, but binding is saturable and only about 86% is bound at usual peak concentration.

Metabolism: Small amounts of roxithromycin are metabolized in the liver and majority of a dose is excreted in the faeces as unchanged drug and metabolites; 53% excreted in the faeces, about 7 to 12% is excreted in the urine and up to 15% via the lungs. Less than 0.5% of an administered dose of roxithromycin is excreted in the breast milk.

Elimination: The elimination half-life is reported to range between about 8 to 13 hours. In children receiving oral roxithromycin 2.5 mg/kg every 12 hours for 5 to 6 days, the elimination half-life is 19 to 21 hours. The increase of this elimination half-life compared to half-life observed in adults is thought to be due to an increased volume of distribution in paediatric patients.

PACK SIZE

Blister pack of 10x10's and 50x10's tablets.

STORAGE CONDITIONS & PHARMACEUTICAL PRECAUTIONS

Store in a cool dry place, below 30°C.

Protect from light.

Keep out of reach of children.

Jauhi daripada kanak-kanak.

Shelf-life: Please refer to the outer box label.

Manufactured by:



SRIPRASIT PHARMA CO., LTD.
216 Moo 6 Suanluang, Krathum Baen,
Samut Sakhon 74110, Thailand.

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